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Prediction of Transition State Structures of General Chemical Reactions via Machine Learning

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Article

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Posted Date: September 29th, 2022

DOI: https://doi.org/10.21203/rs.3.rs-2082595/v1

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Prediction of Transition State Structures of General Chemical Reactions via Machine Learning

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Abstract

The elucidation of transition state (TS) structures is essential for understanding the mechanisms of chemical reactions and exploring reaction networks. Despite advances in computational approaches, TS searches remain still a challenging problem due to the difficulty of constructing an initial structure and heavy computational costs. Herein, a novel machine learning (ML) model for predicting TS structures of general organic reactions is proposed. The proposed model derives interatomic distances of a TS structure from atomic pair features reflecting reactant, product, and linearly interpolated structures. The model shows excellent accuracy, particularly for the atomic pairs where bond formation or breakage occurs. The predicted TS structures result in a high success ratio (93.8%) of quantum chemical saddle-point optimizations, and 88.8% of the optimization results have energy errors of less than 0.1 kcal/mol. Additionally, as a proof-of-concept, exploring multiple reaction paths of an organic reaction is demonstrated with the ML inferences. I envision that the proposed approach will aid the construction of initial geometry for TS optimization and reaction path explorations.

Keywords: Transition state, Machine learning, Reaction path, Chemical reaction

1 Introduction

A transition state(TS) is essential to understanding chemical reactions and reaction networks.[1–3] In principle, the characteristics of chemical reactions are determined not by a specific molecular conformation, but by molecular trajectories that are affected by the entire potential energy surface (PES). Various advanced computational methods have been proposed to effectively simulate the molecular trajectories of chemical reactions.[4–8] However, because of the tremendous computation costs of obtaining accurate PESs and trajectories, elucidations of chemical reactions still rely heavily on classical TS theory.[9, 10] The TS theory is used to derive many reaction properties from a TS, a saddle-point on PES.[11, 12]

Experimentally capturing TS structures is hardly accessible owing to their transient nature. In contrast, they can be accurately determined by computational methods. Over the last five decades, various quantum chemical methods using the first and second derivatives of energy with respect to atomic coordinates have been proposed to accurately capture a saddle-point in PES.[13–16] These methods can be used to reliably determine the TS of target reaction based on a given initial structure. However, they demand precise chemical knowledge on a target reaction to build an initial conformation. Because of the high complexity of PESs and the high sensitivity of quantum chemical calculations to the initial structures, dozens of numerical experiments may be required to design an appropriate initial structure, even for trained chemists. This repetitive trial and error is one of the hurdles to calculate TS structures, along with the high computational cost of quantum chemical methods.

On the other hand, machine learning (ML) techniques are broadly applied in many contexts of chemistry to build chemical knowledge from a wide database. [17–19] By the introduction of various neural network architectures, many molecular and reaction properties (e.g. molecular energy and activation barrier) are accurately predicted.[20–23] Despite those advances of ML techniques, a prediction of 3-dimensional(D) molecular structure is not easily achievable because the inferred 3D structures need to satisfy not only the permutation invariance with respect to the atom order but also physical (rotational and translational invariances) and Euclidean (triangle inequality) constraints.

Recently, researchers have developed ML architectures that can predict TS structures while preserving these conditions. Pattanaik et al.[24] used a graph neural network to generate the initial interatomic distance and weight matrix from interatomic distances of reactant and product structures. This model, referred to as TSGen hereinafter, adopt internal nonlinear optimization to fine the atomic positions with the closet interatomic distance matrix to the initial one. A second model, TSNet was released by Jackson et al.[25] This model is based on a tensor-field network that applies spherical harmonics as convolution filters to distinguish relative atomic positions and directly predicts the atomic positions of TS structures.[26] These ML architectures mathematically preserve the conditions for determining TS structures. To achieve further

improvements on quaility of prediction, a specialized ML model for a specfic reaction type can be introduced.[27] However, the applicable range of specialized ML models are limited to the reaction with the same mechanism. Hence, these models cannot be suitable approach for exploring the chemical reaction space.

Here, a novel ML model that infers TS structures of general single-step reactions is proposed and it is used to find multiple TSs that yield the target reactant and product. Using two organic reaction databases, it is confirmed that the proposed model outperforms existing models particularly for predicting the rarely distributed cases (e.g. bond formation or breakage). In addition to estimating the error of predicted distances, the quantum chemical calculations are performed to measure the chemical validity of the predicted TS structures. The most predicted TS structures converge to saddle-points, and their energies have high agreement with the reference energy. Additionally, by utilizing ML predictions of TS structures, a fast TS finder is implemented.

2 Results

2.1 Prediction accuracy

Schematic representations of the proposed model architecture and its inference procedure are present in Fig. 1. The model is designed to predict the interatomic distances of TS structures from three molecular structures: reactant, product and their linear interpolation. Each structure is featurized as a set of atomic pair features. The pair features of three structures are constructed by concatenating features from two atomic numbers and an interatomic distance (d_{ij}). The sets of pair features from reactant, product and linearly interpolated structures are marked with green, red and purple round boxes. These sets of pair features are updated by pair sequence interaction (PSI) layers without loss of permutation-invariance and size-extensivity. Figure 1 (b) represents PSI layer which consists of two different components; transformer encoder and bidirectional gated recurrent unit (GRU) layers. The transformer encoder updates the pair features by reflecting only a set of pair features belonging to one structure whereas the following bidirectional GRU updates the feature by considering the same pairs in different structures. From the updated features of the interpolated structures($f_{ij}^{\mathrm{I,pair}}$), the model predicts the normalized distances of TS structures by those of the interpolated structure ($d_{ij}^{\mathrm{TS}}/d_{ij}^{\mathrm{I}}$). Further details of the model architecture and training are provided in Section 4.2.

The model outputs are then converted to the predicted interatomic distances (d_{ij}^{TS}) and used to generate the atomic positions in the TS structures (X^*) by nonlinear optimization. As shown in Fig. 1 (c), the nonlinear optimization determines the atomic positions by minimizing the difference between the predicted and reconstructed interatomic distances. To mitigate the effects of prediction errors on the results of nonlinear optimization, the inferences from the trained ML models can be utilized, which named ensemble method. Here,

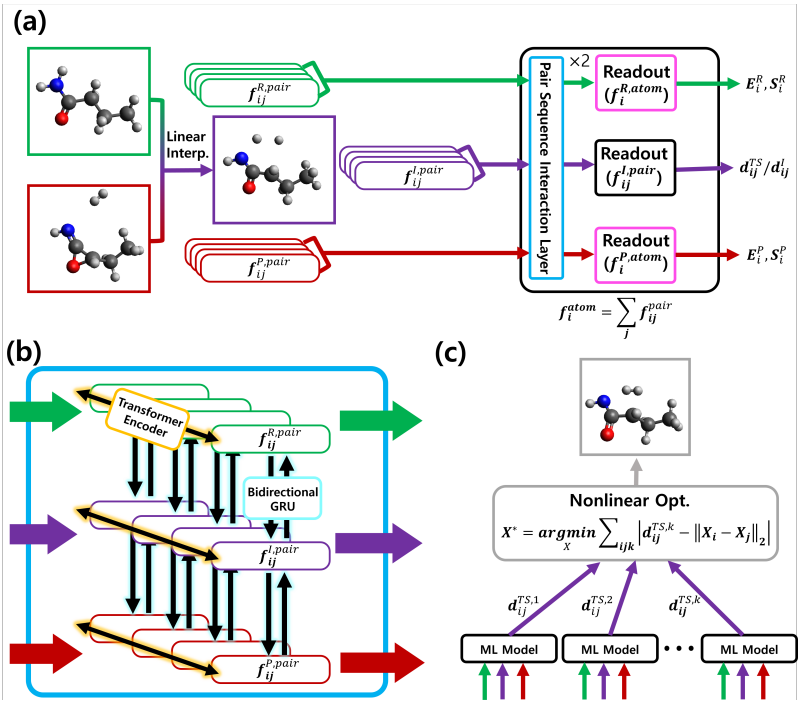

Fig. 1 Schematic representations of machine learning model. (a) Overall machine learning procedure utilizing reactant, product and linearly interpolated structures. **(b)** Illustration of pair sequence interaction layer. **(c)** The predicted interatomic distances are used to reconstruct three-dimensional atomic positions by nonlinear optimization. For ensemble predictions, results from multiple models can be utilized in a single nonlinear optimization.

90 trained models are employed for ensemble predictions. For both single-model and ensemble cases, the linearly interpolated structure is used for a initial geometry of the nonlinear optimization. The details of the ensemble predictions are explained in Section 4.5.

The proposed model is trained and validated using the organic reaction database released by Grambow et al.[28] Figure 2 plots the test errors of the single-model and ensemble prediction results in terms of two metrics: molecular mean absolute error (MAE) and molecular mean absolute percentage error(MAPE). Molecular MAE is appropriate for estimating overall deviations of bond distances. However, in terms of chemical aspects, errors in interatomic distances are not fairly important; while relatively large errors can be tolerated at sufficiently long interatomic distances, even small errors can be critical to the TS structure if they occur in the chemical bonding region. Therefore, to sensitively reflect errors at small distances, a new metric, molecular MAPE, is introduced. The mathematical definitions of both metrics are present in Section 4.3.

The error of the outputs of the single-model and ensemble include 3.681% and 3.407% (molecular MAPE), respectively, and 11.56pm and 10.70pm

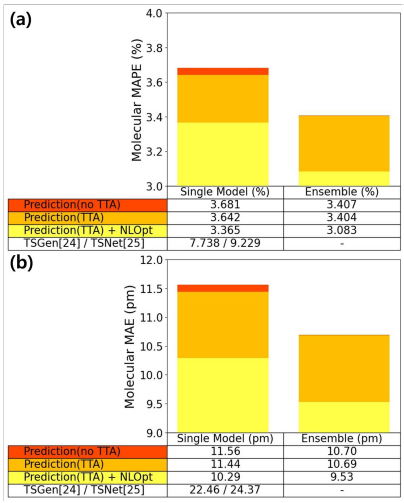

Fig. 2 Accuracy of trained models for predicting interatomic distances. (a) Molecular mean absolute error (MAE) and **(b)** molecular mean absolute percentage error (MAPE) of single-model, ensemble, and comparison model results. The red, orange, and yellow bars represent the magnitudes of errors from direct inferences, test-time augmentation(TTA), and TTA with nonlinear optimization (NLOpt). The last row in each table lists the test errors of ML models proposed by other groups. Both comparison models are trained with the same database as the database used for training of the proposed model.

(molecular MAE), respectively. The error is reduced further by test-time augmentation (TTA), which is a useful ML technique for mitigating the variance of inference without additional training costs.[29, 30] In this study, the augmentation of chemical reactions is performed by reversing the their direction, and the inference results from the forward and backward reactions are averaged. This equates to averaging the TS structures of the forward and backward reactions, which should be identical from a chemical perspective. Thus, in addition to the improvement of prediction accuracy, TTA eliminates order-dependence of outputs according to the direction of a chemical reaction.

The interatomic distances obtained from ML inferences are highly accurate; however, they do not directly correspond to reliable 3D TS structures. Fortunately, by nonlinear optimization using the results of inferences, accurate molecular geometries whose errors in the interatomic distances are less than those of both single-model and ensemble results are obtained. This indicates that the remaining error on the predicted distances can be mitigated by constraining a set of interatomic distances to satisfy the Euclidean condition.

To compare the performances against the other ML models, two existing models, TSGen and TSNet, are trained with the same train dataset.[24, 25]. TSGen exhibited slightly better performance than TSNet. However, both models showed more than two-fold greater errors (for both error metrics) compared with the proposed model.

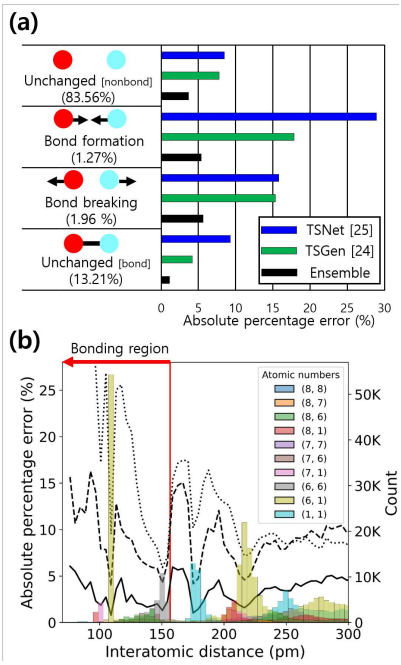

Fig. 3 Error analysis of machine learning predictions. (a) Average values of absolute percentage errors of four interatomic pairs depending on bond change types during a reaction. Blue, green, and black bars represent the prediction errors from TSNet, TSGen and the ensemble approach, respectively. The percentages in parentheses represent the ratio of the type of interatomic pair in the entire training set. The presence and absence of chemical bond in reactant and product is simplified as whether an interatomic distance is less than 156.6pm or not. **(b)** Absolute percentage errors and numbers of atomic pairs with different element sets according to a length of interatomic distance. The dotted, dashed, and solid lines show the average error of TSNet, TSGen and the ensemble predictions, respectively. The bars represent the frequencies of distances in the training set by atomic numbers. The red line indicates 156.6pm, which is criteria for the presence of a chemical bond.

For detailed error analysis, Fig. 3 (a) plots the averages of the absolute percentage error for four different categories of atomic pairs. The first category, which is the most common in the database, corresponds to atomic pairs that are not bonded in either the reactant or product. The second and third categories correspond to atomic pairs that undergo bond formation and bond breakage, respectively. The last category corresponds to the atomic pairs that are bonded in both the reactant and product. Because only a limited number of bond changes occurs in single-step reactions, the second and third cases are relatively rare (1.27% and 1.96% of all interatomic pairs in the training set, respectively). Despite the limited number of the pairs belonging to the second and third categories, the ensemble predictions achieve ∼5% errors on the both categories while two comparison models record more than 15% errors. In the prediction of TS structures, determining interatomic distances that induce

chemical bond formation and breakage are essential. The comparison models failed significantly in these cases.

To observe the dependency of the model accuracy on the size of the interatomic distances and atomic numbers, changes of absolute percentage errors and frequncies of atomic pairs in the database are shown in Fig. 3 (b). The colored bars represent the distributions of ten different combinations of atomic pairs at intervals of 4.125pm. The dotted, dashed, and solid black lines show the absolute percentage errors of the interatomic distances using TSNet, TSGen, and the ensemble approach. The points at the shown lines are obtained by averaging the percentage errors of all training interatomic distances belonging to the same interval to the bars. The red line represents the criteria for chemical bond, 156.6pm, which is used for determining the category of each atomic pair.

In Fig. 3 (b), large absolute percentage errors are commonly observed in the three ML predictions where interatomic distance are rarely distributed. Especially, in the bonding region, all models record relatively high error except around 105pm and 150pm regions in which a large number of C–H and C–C bonds are distributed. Despite this common trend, the proposed model outperforms both TSNet and TSGen throughout the entire distance range. Even for the atomic pairs in the bonding region, at most 6% of errors are observed in the ensemble prediction while the comparison models record up to 15% or more errors. This analysis confirms that the proposed model provides reliable predictions for all types of atomic pairs, particularly for those atomic pairs that have chemical bond bonding and participate reactions. In addition, it has a high accuracy for atomic pairs consisting of infrequently distributed elements.

To investigate the applicability of the proposed approach, the model is trained using the S_N2 database, which was published by the authors of TSNet.[25] For S_N2 reactions, the proposed ensemble approach recorded 4.54pm (molecular MAE) and 1.738% (molecular MAPE). For the S_N2 reaction database, the best-reported MAE is 18.31 pm using the TSNet model.[25] Because the used definition of error is slightly different from the molecular MAE, if the same definition is applied, the error of the ensemble results becomes 4.97pm. Consequently, the proposed model shows is more than three times as accurate as the TSNet model. In fact, the model based on PSI layers achieved overwhelming accuracy in both experiments. The proposed model utilize linearly interpolated structure in two aspects; normalization of interatomic distances for output values and construction of input features. As aforementioned, the interpolated structure provides reliable interatomic distances for rarely distributed cases so that the proposed model can result in the refined interatmoic distances for those rare atomic pairs.

To estimate chemical validity of the predicted TS structures, quantum chemical simulations are performed. Although the S_N2 database includes many realistic chemical reactions, the validation based on quantum chemical calculations is only conducted for only the database released by Grambow et al.[28] because of the small size of S_N2 database.

2.2 Quantum chemical validation

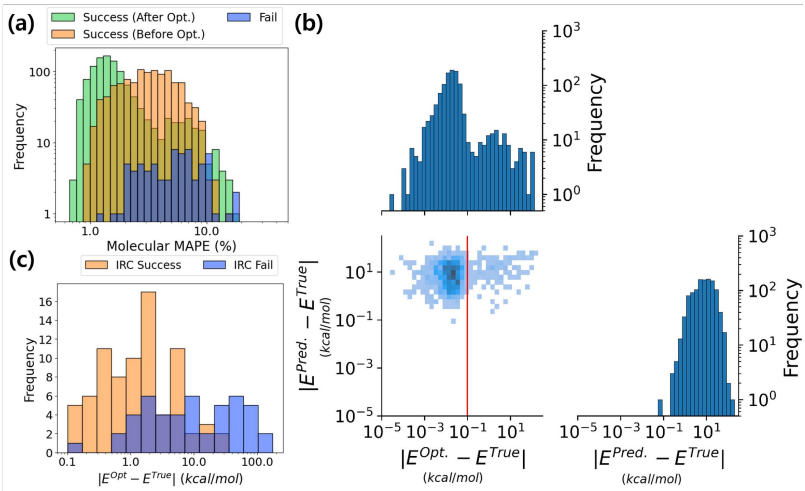

Fig. 4 Quantum chemical validation results of predicted transition state structures (a) Distributions of molecular MAPE of inferred and quantum chemically optimized TS structures. Orange and blue bars indicate the molecular MAPE of the structures that yielded successful and unsuccessful convergence in the saddle-point optimizations, respectively. Green bars indicate molecular MAPE of optimized structures. **(b)** Distributions of absolute energy error of inferred and optimized structures. The red line indicates an absolute energy error of 0.1kcal/mol. Intrinsic reaction coordinate (IRC) calculations are performed for the structures on right side of the red bar (126 cases). **(c)** Distributions of absolute energy errors of structures whose reference product and reactant structures are acquired (orange) or not acquired (blue) by IRC calculations.

To validate the ML results, saddle-point optimizations are performed using the predicted TS structures as the initial structures. Among the 1196 test molecules, 1122 (93.8%) of the molecular structures are successfully converged. This high success ratio implies that the model inferences provide structures close to the reference ones. Of the failed saddle-point optimizations, 60 failed because the maximum number of iterations for geometry relaxation is exceeded; six failed because self-consistent field failed to converge; and eight failed for other reasons.

The effect of saddle-point optimization on TS structures is evaluated by plotting the distributions of molecular MAPE before and after the saddle-point optimization, as shown by the orange and green bars in Fig. 4 (a), respectively. The MAPE distribution shifted to the left after saddle-point optimization, which corresponds to a reduction of the error. However, a considerable portion of the structures remained on around 8% molecular MAPE, creating a bimodal distribution. The splitting of distribution due to quantum chemical refinements is also observed in the energy error. Figure 4 (b) shows the distribution of the absolute energy error in the optimization results. The absolute energy errors of the predicted structures are broadly distributed at 0.1–100 kcal/mol.

However, for the optimized structures, the absolute energy errors of most of the structures (996 cases) are reduced to below 0.1 kcal/mol, and only a small number of cases remained within the original error range. This implies two things. First, the energy of the inferred structure is not sufficiently accurate, even if the inferred interatomic distances have an error of only few percentage on average. Second, quantum chemical optimization reduces energy errors and yields reliable energy in most cases; however, in some cases, the results of saddle-point optimization are still far from the reference TS.

Intrinsic reaction coordination (IRC) calculations are performed to verify whether the reaction paths with the optimized TS structures are appropriately connected to the reference reactant and product structures. The structures with errors greater than 0.1kcal/mol (shown on the right-hand side of the red line in Fig. 4 (b)) are selected for testing (126 cases). If the reference reactant and product structures are obtained by IRC calculations, the cases are classified into the IRC success group, otherwise, they are classified into the IRC failure group. The orange and blue bars in Fig. 4(c) represent the energy error ($|E^{\mathrm{Opt}} - E^{\mathrm{Ref}}|$) distribution of the IRC success and IRC fail groups, respectively. Generally, the IRC success cases (79 structures) have lower $|E^{\mathrm{Opt}} - E^{\mathrm{Ref}}|$ values than the IRC failure cases. In particular, in the range below 0.5kcal/mol, only one structure failed to reproduce the reactant and product structures. Conversely, even some structures with a difference of more than 10kcal/mol successfully yielded the correct reactants and products. An energy difference over 10kcal/mol indicates that the optimized TS structure has a significantly different conformation from the reference structure. Despite this large discordance, these cases can be included in the IRC success group because of the nonuniqueness of reaction paths.

Figure 5 visualizes two extreme IRC success cases with respect to the energy error. The upper and lower subplots show the reference, predicted, and optimized TS structures for the cases with the most positive and most negative energy errors, respectively. In both positive and negative error cases, it is observed that the predicted TS structures are not significantly changed by the saddle-point optimization while the reference TS structures are noticeably different from both the predicted and optimized TS structures. This means that the proposed model predicts one of the TS structures, and it is well optimized to the closest TS structure, but it is not the same as the reference one. In the positive error case, the activation energy of the reaction passing through reference TS (E_a^{ref}) is lower than that of the optimized TS (E_a^{opt}) because the energy of optimized structure is higher than that of reference one. In contrast, the negative error indicates that E_a^{opt} is less than E_a^{ref} which means that the ML method predicts energetically more favorable TS structure than reference one.

Multiple TSs for a chemical reaction are frequently observed in general polyatomic systems. In a strict sense, it is extremely difficult to precisely target the most stable TS structure; however, practically, the most stable TS structure can be found by exploring multiple TSs. Therefore, the generation

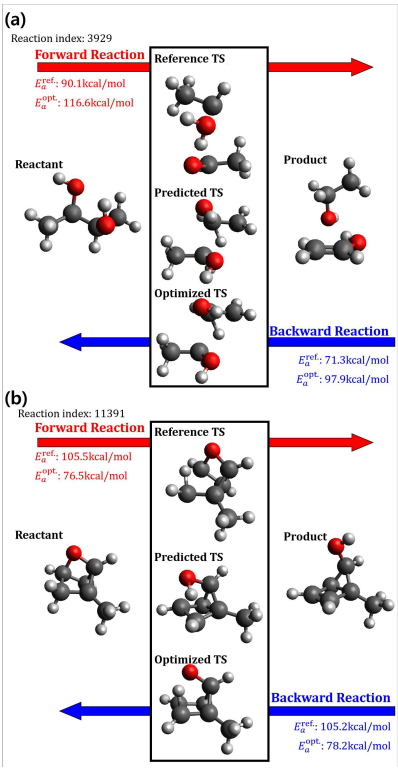

Fig. 5 Molecular structures and activation barriers for extreme cases Reactant, product and transition states having **(a)** the most positive and **(b)** the most negative energy errors among the inferred structures that yielded correct reactant and product structures through intrinsic reaction coordinate calculations. A positive (or negative) error indicates that the obtained TS structure is energetically less (or more) probable than the corresponding reference transition state.

of multiple reaction pathways is essential for many applications. Furthermore, to design catalysts or retrosynthetic pathways, acquiring information on competitive reaction mechanisms is a challenging but demanding work.[31–33] To tackle this problem, generation of multiple TS structures using the fast and accurate ML inferences is demonstrated with the aid of reactant and product sampling.

2.3 Exploring multiple reaction paths

For the reaction in which the ML model fails the most (Fig. 5 (a)), multiple TS structures are found by using the ensemble approach with normal mode sampling (NMS) of reactant and product. NMS can quickly sample a large number of thermally activated structures because it perturbs atomic positions along the normal modes of the equilibrium structure instead of heavy molecular dynamic simulations.[34] From the 2000 sets of sampled reactants and products, the same number of TS structures are generated by the ensemble method.

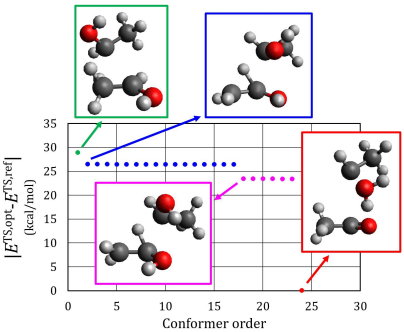

Fig. 6 Relative energies from transition state exploration The TS exploration is performed for the reaction shown in Fig. 5 (a) using quantum chemical saddle-point optimization and the proposed machine learning approach with off-equilibrium samplings of reactant and product. The relative energy of 24 TS structures from the exploration are plotted, in descending order, with respect to the energy of reference TS structure, $E^{\mathrm{TS,opt}} - E^{\mathrm{TS,ref}}$. The energies with the same color share the same conformation. The representative structures are shown in boxes of the same color.

Generating 2000 TS structures typically requires considerable human labor or computational cost, even if an automatic reaction path finder is applied. In contrast, only 1 minute is spent for inferring 2000 TS structures with NVIDIA Titan RTX graphic card. Although 90 ML models are used for the ensemble predictions, these inferences can be performed independently. Therefore, the computational expense of even ensemble inferences is manageable. To reduce computational costs of quantum chemical refinement, instead of simulating all generated TS structures, 117 representative TS structures are selected by clustering and they are used for saddle-point calculations as initial geometries. (See Section 4.6 for details the NMS and clustering procedures) Among the 117 calculations, 24 calculations successfully converged.

Figure 6 shows the relative energies ($|E^{\mathrm{TS,opt}} - E^{\mathrm{TS,ref}}|$) of the 24 successfully converged TS structures relative to the reference structures. Among 24 structures, four conformations are observed. The energy deviations among structures with the same conformations are less than 0.05kcal/mol. The energies of each conformations and its corresponding structure are shown with the same color. The conformation with the lowest energy (shown in the red box) is the same as that of the reference TS. The conformations within blue and pink boxes share most structural characteristics, except for the direction of the –OH group. In contrast, the conformation shown in the green box has a completely different structure even though its energy is close to those of other two conformations. By utilizing ensemble inferences with NMS, four different TS structures are obtained within a reasonable number of quantum chemical calculations.

To validate the TS structures, IRC calculations are performed to the obtained four conformers shown in Fig. 6. It is confirmed that reference reactant and product are obtained from IRC calculations using the structures shown in green, blue, and pink boxes as an initial geometry. In contrast, the

structure in the red box, which has the same conformation to the reference TS, does not yield the correct product structure. Instead of the reference product, the IRC calculation generates the conformation consisting of three molecules (C_2H_5, H_2O, and C_2OH_3). Interestingly, this same products are generated by IRC validation of the reference TS structure. Because the reference TS structures are generated from the single-ended growing string method followed by refinement with saddle-point optimization, the obtained TS can be transformed into the TS of unwanted reaction during the saddle-point optimization. Although one of the obtained TS is not a correct TS of target reaction, these results confirm that the proposed method can generate reliable TS structures, even with off-equilibrium reactant and product structures. Additionally, the effectiveness of the proposed approach for tackling multiple reaction paths is also demonstrated.

The IRC validations of the obtained TS conformations reveals that the large energy errors of the originally predicted TS structures may not be actual errors of the model, but the reference database. To systematically evaluate data errors in the TS database, a large number of IRC calculations are required, which is not viable using current computing resources. Despite some potential faults in the reference structures, the proposed ML model is stably trained and can be applied to exploring multiple reaction paths which is a challenging problem in many contexts of chemistry.

3 Discussion

In this study, a novel ML model for fast and accurate prediction of TS structures is proposed and validated using general organic chemical reactions. To achieve high accuracy with the satisfaction of various necessary invariances, the ML model contained PSI layers designed to handle any size of molecules and any type of element. This layer relies not on any prior knowledge of chemical reaction type but on reactant and product structures, thus can be extended to all types of chemical reactions without modifications. Despite the lack of reliable TS data, the proposed ML model shows high accuracy for general organic reactions. In particular, for atomic pairs that undergo bond formation or breakage during the reaction, the proposed model noticeably outperforms existing models. Additionally, when the inferred TS structures are used as inputs for quantum chemical saddle-point optimizations, a high convergence ratio is achieved, which demonstrates that this ML method can be an attractive solution for initializing a starting structure for saddle-point optimizations.

To demonstrate the potential usage of the model in searching multiple reaction paths, an autonomous reaction path finder is implemented and tested on a polyatomic reaction. By performing ML inference from the off-equilibrium conformations of reactant and the product and further refinement by saddle-point calculations, various chemically meaningful TS structures are identified. This automated TS finder has huge potential to leverage ML applications for chemical reactions because it can be directly utilized for not only chemical

applications (e.g. design of synthetic routes and catalysts), but also various ML applications for chemical reactions (e.g. active learning). Hopefully, the proposed ML approach as well as its components (such as TTA, and ensemble) will be utilized to construct powerful artificial intelligence in the field of chemical reactions.

4 Methods

4.1 Data preparation and quantum chemical validation

In this study, two publicly available reaction databases are used. The first was released by Grambow et al.[28] This database is consist of density functional theory (DFT) results with two different options, B97-D3/def2-mSVP and ωB97X-D3/def2-TZVP. Here, ωB97X-D3/def2-TZVP results (11961 reactions) are employed because of its reliable accuracy. To extract atomic positions and molecular properties from the DFT results, *ard_gsm* package provided by the authors of the database is utilized. (www.github.com/cgrambow/ard_gsm) The database, is split randomly into training, validation, and test sets with an 80-10-10 ratio. The second database includes 53 S_N2 reactions which was released by the author of TSNet.[25] This database includes five validation and 48 training data. The same splitting result is adopted.

Saddle-point optimization and IRC calculations are performed for the test reactions in the first database using the same DFT functional(ωB97X-D3) and basis set (def2-TZVP). Saddle-point optimization is performed with TS optimization implemented in QChem while pysisyphus program is utilized for IRC calculations.[35] The pysisyphus program provides various methods of exploring PESs based on the results (mainly, gradients and Hessian) of a quantum chemical package. To run IRC calculations with pysisphus, Psi4 package is employed because it is seamlessly coupled to pysisyphus and supports ωB97X-D3 and def2-TZVP.[36]

4.2 Model Architecture and Training

Linear interpolated geometries are frequently used for TS search algorithms because of their reliable structures and low costs. [13, 37] The proposed model predicts the interatomic distances of the true TS from the reactant, product, and linearly interpolated structures. These three structures are featurized as a sequence of atomic pair sets. The pair features are constructed by concatenating one distance and two atomic features. For atomic and distance features, entity embedding of atomic numbers and Gaussian kernel are employed, respectively. For centers of Gaussian functions, points on an equidistant grid are used. The information on the grid is present in Table S1. Each structure includes a set of pair features (f_{ij}) and reactions are featurized as a sequence of structure feature sets whose dimension is $(\frac{N_{\text{atom}} \times (N_{\text{atom}}+1)}{2}, 3, N_f)$ where N_{atom} and N_f are number of atoms and feature size for the pairs, respectively. For convenience, in Fig. 1, two atomic indices (i, j) are introduced; however, only one index is

used for atomic pairs in the actual implementation. The pair features satisfy the translational and rotational invariances because neither atomic numbers nor distances are changed by translation or rotation.

Because the number of atoms participating in a reaction is not fixed, and because their order is not chemically meaningful, operations on the first dimension of pair features need to be size-extensive and permutation-invariant, whereas the size of the second dimension is always fixed and the order is not interchangeable. To preserve the nature of data structure, the novel layer named pair sequence interaction (PSI) layer is introduced.

Initially, the PSI layer updates the pair features by reflecting other pairs in the same structure. This procedure consists of a transformer encoder, which is a permutation-invariant and size-extensive operation. This first update is not affected by the pair features of other structures. Subsequently, the atomic features are updated again to reflect the information from other structures. Because this second update proceeds only among only atomic pairs sharing the same index, the permutation invariance and size-extensivity of the first dimension are still preserved. For the second update, a GRU is used. Figure 1(b) illustrates the two updates of the PSI layer using different colors. The yellow operation operates only on pair features in the same structure, whereas the following blue operation updates pair features by reflecting pairs with the same index but in other structures. By stacking multiple PSI layers, the effects of all the atomic pairs of the reactant, product, and linearly interpolated structures participate in determining every one of the final atomic pair features. In the proposed model, two PSI layers are stacked and the initial feature size of atomic pair feature (128) is changed to 512 at the end of the two stacked PSI layers because each bidirectional GRU doubles the feature size.

To compute the reaction properties from the updated atomic pair features, two readout layers are used. The first of these readout layers, represented as the purple box in Fig. 1 (a), is applied to compute the ratios of the interatomic distances between the TS structure and linearly interpolated structure. The second readout layer computes the contribution of each atom to total energy, rotational entropy, and translational entropy. (E_i,$S_{(vib,i)}$ and, $S_{(rot,i)}$) The atomic features for the second layer is computed using the following equation:

$$f_i^{\text{atom}} = \sum_j f_{ij}^{\text{pair}} \tag{1}$$

where f_i^{atom}, and f_{ij}^{pair} are the atomic and pair features. Because the model only computes f_{ij}^{pair} when $i \geq j$, the omitted pair features are recovered using symmetric approximation, $f_{ij}^{\text{pair}} = f_{ji}^{\text{pair}}$. The predicted values from the second readout layers are utilized for multi-label learning which is frequently used to improve the quality of ML inferences by leveraging the information from the related information.[38, 39]

Implementation and training of the proposed model are performed with PyTorch Lightning, which supports a high-level interface of PyTorch and

is freely available at https://github.com/Lightning-AI/lightning. The implemented model and scripts for training and inference are uploaded at https://gitlab.com/sunghwan.choi/learnts. All hyperparameters used to train the model are summarized in Table S1.

4.3 Metrics

Molecular MAE and molecular MAPE are defined as follows:

$$(\text{Molcular MAE}) = \frac{1}{M}\sum_{t}^{M}\frac{2}{N_{\text{atom},t}(N_{\text{atom},t}-1)}\sum_{i<j}|d_{ij}^{t,\text{true}} - d_{ij}^{t,\text{pred}}| \quad (2)$$

$$(\text{Molcular MAPE}) = \frac{1}{M}\sum_{t}^{M}\frac{2}{N_{\text{atom},t}(N_{\text{atom},t}-1)}\sum_{i<j}\frac{|d_{ij}^{t,\text{true}} - d_{ij}^{t,\text{pred}}|}{d_{ij}^{t,\text{true}}} \quad (3)$$

where $d_{ij}^{t,\text{true}}$ and $d_{ij}^{t,\text{pred}}$ are the interatomic distances between the ith and jth atoms of the geometries for a t-indexed reaction. M and $N_{\text{atom},t}$ are a number of reactions in a database and a number of atoms in the t-indexed reaction, respectively. The distance error used for TSNet is defined in Equation 4 in Ref [25].

4.4 Loss function

During the training, the weighted sum of three losses are minimized. The first loss, L_1, is molecular MAE of the predicted TS interatomic distances. The second loss, L_2, includes the error in reaction properties. Here, the changes in total energy and entropies during reactions participate to evaluate the second loss. The atomic contributions of molecular properties are evaluated from the second readout layer, thus the outputs are summed along the atom index to evaluate molecular properties of each structure. L_2 is evaluated as the sum of differences of three reaction properties (total energy, rotational, and translational entropy changes) between predicted and reference values. Because, reference entropy values are not available in the S_N2 database, the second loss consists of only the energy changes. The last loss, L_3, represents the constraints of the Euclidean distance matrix whose elements are the squares of interatomic distances. In addition to the hollow and symmetric conditions, a Euclidean distance matrix should satisfy the eigenvalue condition, which is nontrivial to implement. The 3D Euclidean distance matrices can have at most 5 nonzero eigenvalues and the the sum of eigenvalues should be zero.[40] To implement both conditions, L_3 is designed as follows:

$$L_3 = \sum_{i=1}^{\min(5,N_{\text{atom}})} e_i + \sum_{i=\min(5,N_{\text{atom}})}^{N_{\text{atom}}} |e_i| \quad (4)$$

where e_i is ith eigenvalue in descending order of magnitude. The first and second term constrains the sum of first five eigenvalues and the rest eigenvalues to zero. By penalizing a distance matrix that cannot realize 3D atomic positions, results of the proposed model are guided to satisfy Euclidean constraint.

The total training loss, L is defined as

$$L = cL_1 + c'(L_2 + L_3) \tag{5}$$

The selection of coefficients are not significant except for large c'. The accuracies of the model depending on the coefficients are summarized in Table S2.

4.5 Nonlinear optimization and Ensemble

Previous studies have used ML models to directly infer the 3D atomic positions.[24, 25] However, here, the ML model predicts only the interatomic distances. Therefore, an additional process is required to reconstruct 3D atomic positions. From the initial atomic positions, X, the nonlinear optimization finds optimal positions, X^*, to minimize the difference to the ML predictions as follows:

$$X^* = \underset{X}{\operatorname{argmin}} \sum_k \sum_{i,j}^{N_{\text{atom}}} w_{ij} |d_{ij}^k - \|X_i - X_j\|_2| \tag{6}$$

where d_{ij}^k and w_{ij} are the predicted interatomic distance from the kth model in the ensemble and the weight factor, respectively. To increase the contribution from short interatomic distances, weight factors are set as Gaussian form, $e^{-\alpha d_{ij}^2}$. By testing many α values on the validation results, optimal $\alpha (= 0.2 \text{Å}^{-2})$ is adopted and used for the test set.

For the ensemble predictions, 90 trained models are employed. Six different combinations of c and c' in Equation 5 are used to train the model and three independent training with different random seeds are performed for every combination of c and c'. Although all the hyperparameters are identical, depending on the random seed, ML model are differently trained. For each training, five best models are saved and used for the ensemble predictions.

4.6 Normal mode sampling and clustering

For each reactant and product, 2000 geometries are sampled using the NMS.[41] The atomic structures of the sampled reactant and product structures are rotated and shifted to minimize the root-mean-square-deviation between the two structures. From the obtained 2000 sets of reactant and product structures, the TS structures are predicted based on the proposed ensemble inferences. To avoid duplicated TS structures, the 117 representative structures are selected using the affinity propagation clustering method implemented in scikit-learn.[42, 43] The features for the clustering is set as interatomic distances.

Supplementary information. The used hyperparameters and accuracies of single-model predictions depending on c and c' are summarized in Table S1 and Table S2, respectively.

Acknowledgements. The authors thank Y. Kim for the helpful discussions. This work has been carried out with the computing resource supporting by KISTI[KSC-2020-CRE-0117].

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